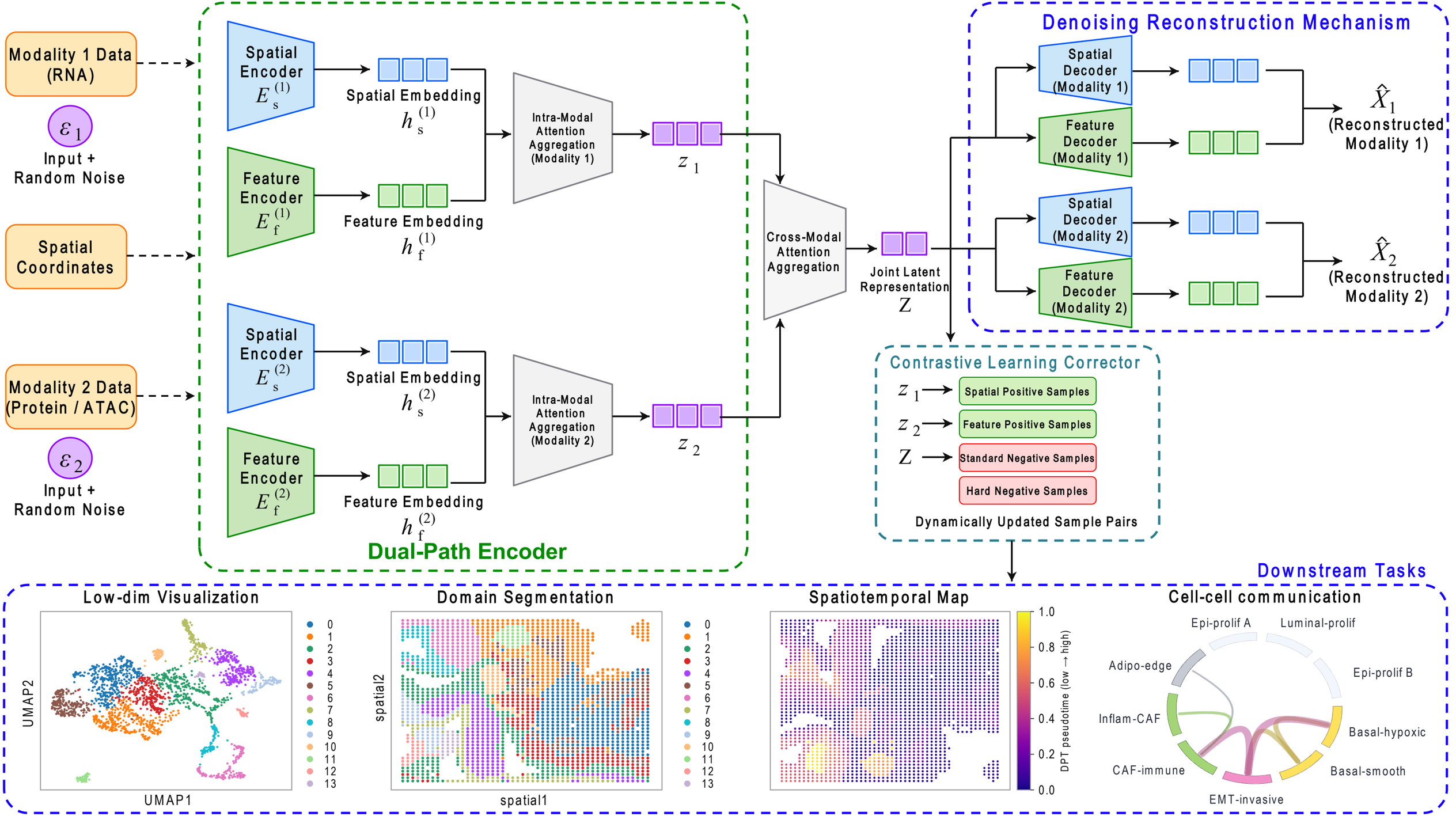

Modality 1 Data (RNA)
ε_1
Input + Random Noise
Spatial Coordinates
Modality 2 Data (Protein / ATAC)
ε_2
Input + Random Noise
Spatial Encoder $E_s^{(1)}$
Feature Encoder $E_f^{(1)}$
Spatial Embedding $h_s^{(1)}$
Feature Embedding $h_f^{(1)}$
Intra-Modal Attention Aggregation (Modality 1)
z_1
Spatial Encoder $E_s^{(2)}$
Feature Encoder $E_f^{(2)}$
Spatial Embedding $h_s^{(2)}$
Feature Embedding $h_f^{(2)}$
Intra-Modal Attention Aggregation (Modality 2)
z_2
Dual-Path Encoder
Cross-Modal Attention Aggregation
Joint Latent Representation Z
Denoising Reconstruction Mechanism
Spatial Decoder (Modality 1)
Feature Decoder (Modality 1)
$\hat{X}_1$ (Reconstructed Modality 1)
Spatial Decoder (Modality 2)
Feature Decoder (Modality 2)
$\hat{X}_2$ (Reconstructed Modality 2)
Contrastive Learning Corrector
z_1 → Spatial Positive Samples
z_2 → Feature Positive Samples
Z → Standard Negative Samples
Hard Negative Samples
Dynamically Updated Sample Pairs
Downstream Tasks
Low-dim Visualization
UMAP2
UMAP1
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Domain Segmentation
spatial2
spatial1
0 1 2 3 4 5 6 7 8 9 10 11 12 13
Spatiotemporal Map
DPT pseudotime (low → high)
1.0 0.8 0.6 0.4 0.2 0.0
Cell-cell communication
Epi-prolif A
Luminal-prolif
Adipo-edge
Epi-prolif B
Inflam-CAF
Basal-hypoxic
CAF-immune
Basal-smooth
EMT-invasive